

Experiment 1 – Simulation Report

Smart Motion and Accident Detection System

1. Objective

To verify the RTL behavior of the Smart Accident Detection System using Questa simulation. The design monitors motion, impact, and tilt inputs and asserts the accident output when both impact and tilt are HIGH.

2. RTL Design Under Test

Module: Smart_Accident_Detection

Inputs: motion, impact, tilt

Outputs: motion_led, impact_led, accident_led

Logic: motion_led = motion; impact_led = impact; accident_led = impact AND tilt.

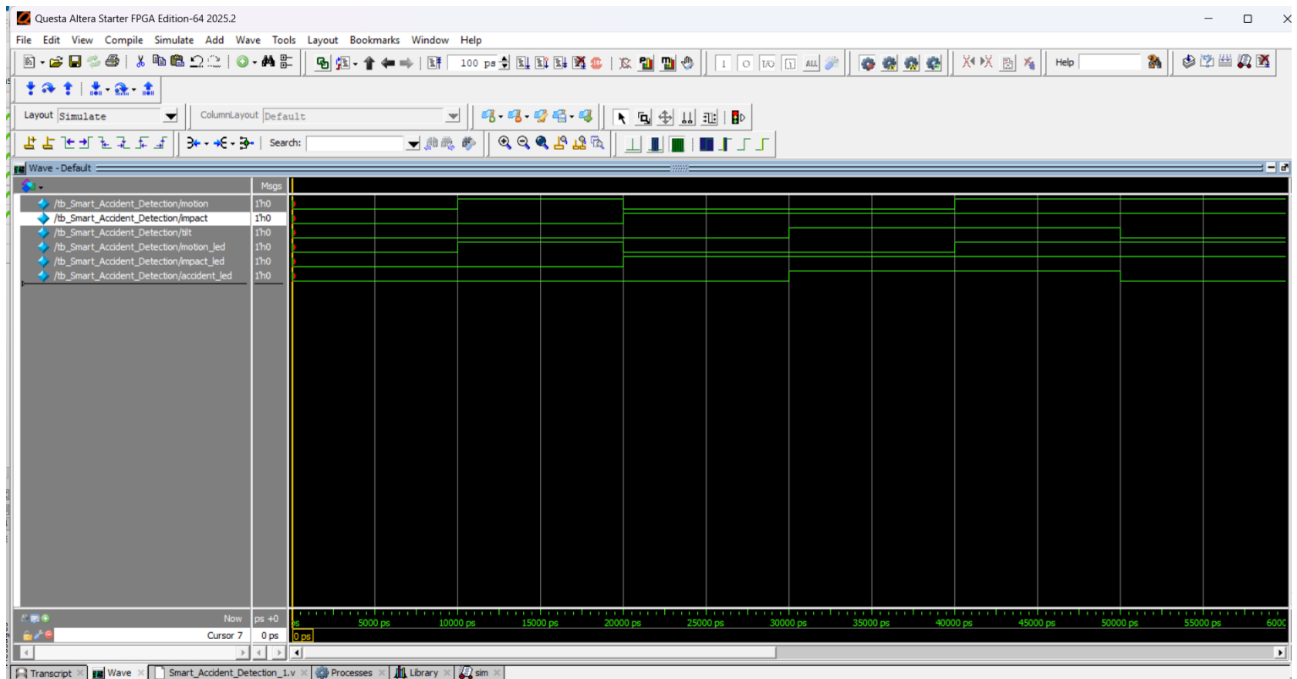
3. Simulation Environment

Item	Result
Simulator	Questa Altera Starter FPGA Edition 64 2025.2
Testbench	tb_Smart_Accident_Detection
Simulation duration	60 ns
Loading / execution	Successful
Simulation errors	0
Warning	1 WLF file-in-use warning; simulation completed successfully

4. Test Sequence and Expected Behavior

Time	motion	impact	tilt	motion_led	impact_led	accident_led
0–10 ns	0	0	0	0	0	0
10–20 ns	1	0	0	1	0	0
20–30 ns	0	1	0	0	1	0
30–40 ns	0	1	1	0	1	1
40–50 ns	1	1	1	1	1	1
50–60 ns	1	1	0	1	1	0

5. Simulation Waveform



The waveform shows the six required signals and their transitions over the 0–60 ns test sequence.

6. Transcript Verification

The Questa transcript records successful loading of work.tb_Smart_Accident_Detection and execution of **run 60 ns**. The simulation stopped at 60 ns at the testbench \$stop statement. The transcript reported zero simulation errors.

7. Conclusion

The Smart Accident Detection RTL was successfully verified in Questa. The motion and impact indicators follow their corresponding inputs, and the accident indicator becomes HIGH only when impact and tilt are simultaneously HIGH. The simulation evidence is suitable for the Experiment 1 submission.